WORKED SOLUTION

STEP II 2024, Question 10

Mechanics • Friction on Accelerating Wedge

STEP II 2024, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

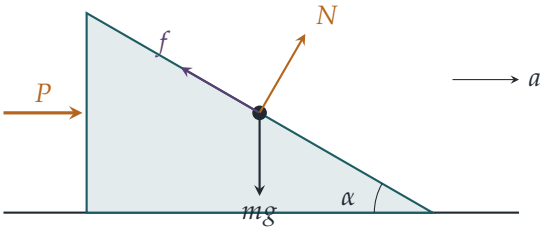
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The key idea

Everything hinges on two observations. First, the plane is smooth, so the only horizontal external force on the whole system (prism plus particle) is P: that gives the common acceleration at once. Second, since the particle does not move relative to the prism, it has exactly that same horizontal acceleration, so resolving for the particle alone along and perpendicular to the slope gives the friction and the normal reaction directly. The inequality in (iii) is then just "friction $\leq \mu\times$ normal reaction", tidied up with the compound-angle formulae.

Solution

We model the particle as a particle, the prism as a rigid body sliding on a smooth horizontal plane, and the contact between particle and oblique face as rough with coefficient μ. Place the vertical face on the left, so that the oblique face slopes down to the right; then P pushes the prism to the right.



On the particle act its weight mg, the normal reaction N perpendicular to the oblique face, and friction, which we take as a signed quantity f acting *up* the slope (if f turns out negative, the friction actually acts down the slope). The magnitude of the friction is $F = |f|$.

Part (i)

Suppose the system is in equilibrium, so everything is at rest. Resolving for the particle perpendicular to and along the slope,

$$N = mg\cos\alpha, \qquad f = mg\sin\alpha.$$

For the particle not to slip we need $|f| \le \mu N$, that is $mg\sin\alpha \le \mu\, mg\cos\alpha$, or $\mu \ge \tan\alpha$ (we can divide by $mg\cos\alpha > 0$ since α is acute). So if $\mu < \tan\alpha$ no equilibrium is possible: the particle must slide.

Part (ii)

The acceleration. Take the prism and particle together. The plane is smooth, so it exerts only a vertical force; the forces between particle and prism are internal. Hence the only horizontal external force is P, and since both bodies share the same acceleration a (horizontal, to the right),

$$P = (M+m)a, \qquad a = \frac{P}{M+m}.$$

Resolving for the particle. The particle's acceleration is a horizontally. Its component up the slope is $-a\cos\alpha$ (the slope rises to the left, and the particle accelerates to the right), and its component along the outward normal is $a\sin\alpha$.

> **Ask yourself** Why resolve along and perpendicular to the slope rather than horizontally and vertically?

Along the slope only f and the weight have components, and perpendicular to it only N and the weight. So each equation contains just one unknown, and we avoid solving simultaneous equations. The price is that we must resolve the acceleration too, which is easy.

Up the slope:

$$f - mg\sin\alpha = -ma\cos\alpha \quad\Longrightarrow\quad f = m(g\sin\alpha - a\cos\alpha).$$

Substituting $a = P/(M+m)$,

$$f = \frac{m}{M+m}[(M+m)g\sin\alpha - P\cos\alpha],$$

and so the magnitude of the friction is

$$F = \frac{m}{M+m}\,|(M+m)g\sin\alpha - P\cos\alpha|,$$

as required. The modulus is genuinely needed: for small P friction acts up the slope, but for large P the slope is pushed into the particle hard enough that friction must act down the slope to stop it riding up.

Perpendicular to the slope (outward normal):

$$N - mg\cos\alpha = ma\sin\alpha \quad \Longrightarrow \quad N = \frac{m}{M+m}[(M+m)g\cos\alpha + P\sin\alpha].$$

This is positive, since $P > 0$ and α is acute, so no modulus is needed and the particle stays in contact.

Part (iii)

The particle stays fixed relative to the prism exactly when friction can supply what is needed: $F \le \mu N$. Write $G = (M+m)g$ and cancel the positive factor $m/(M+m)$:

$$|G\sin\alpha - P\cos\alpha| \le \tan\lambda\,(G\cos\alpha + P\sin\alpha).$$

Multiplying by $\cos\lambda > 0$, this is the pair of inequalities

$$-\sin\lambda\,(G\cos\alpha + P\sin\alpha) \;\le\; \cos\lambda\,(G\sin\alpha - P\cos\alpha) \;\le\; \sin\lambda\,(G\cos\alpha + P\sin\alpha).$$

Ask yourself How do these turn into tangents of $\alpha \pm \lambda$?

Collect the G terms on one side and the P terms on the other; each bracket is then a compound-angle expansion.

Right-hand inequality (friction up the slope must not exceed its limit):

$$G(\sin\alpha\cos\lambda - \cos\alpha\sin\lambda) \le P(\cos\alpha\cos\lambda + \sin\alpha\sin\lambda),$$
$$G\sin(\alpha-\lambda) \le P\cos(\alpha-\lambda).$$

Since $0 < \alpha - \lambda < \frac{1}{4}\pi$, $\cos(\alpha-\lambda) > 0$, so this is $P \ge G\tan(\alpha-\lambda)$.

Left-hand inequality (friction down the slope must not exceed its limit):

$$P(\cos\alpha\cos\lambda - \sin\alpha\sin\lambda) \le G(\sin\alpha\cos\lambda + \cos\alpha\sin\lambda),$$
$$P\cos(\alpha+\lambda) \le G\sin(\alpha+\lambda).$$

Here we use the condition $\lambda < \alpha < \frac{1}{4}\pi$: it gives $0 < \alpha + \lambda < \frac{1}{2}\pi$, so $\cos(\alpha+\lambda) > 0$ and we may divide to get $P \le G\tan(\alpha+\lambda)$.

Both must hold, so

$$(M+m)g\tan(\alpha-\lambda) \le P \le (M+m)g\tan(\alpha+\lambda),$$

as required.

Checks

- With $P = 0$ the lower bound says $0 \ge G\tan(\alpha-\lambda)$, which is false because $\lambda < \alpha$: consistent with part (i), since $\mu = \tan\lambda < \tan\alpha$.

- At $P = G\tan\alpha$ we get $f = 0$: the push is exactly right for a smooth particle to sit still, which is the familiar "accelerating wedge" result $a = g\tan\alpha$. This value lies inside the interval, as it should.
- Numerically, with $M = 3$, $m = 1$, $g = 9.8$, $\alpha = 30^\circ$, $\lambda = 10^\circ$, the allowed range is $14.27 \le P \le 32.89$ (newtons). At $P = 20$ the formulae give $F \approx 0.570$ and $N \approx 10.99$, so $F/N \approx 0.052 < \tan 10^\circ \approx 0.176$. Solving the particle's equations directly for thousands of random cases by computer agrees with both formulae and with the interval.

Common slips

- Taking $N = mg\cos\alpha$ in part (ii): the slope is accelerating, so the normal reaction is not the static one.
- Dropping the modulus and finding only the lower bound on P; the upper bound comes from friction acting *down* the slope.
- Dividing by $\cos(\alpha + \lambda)$ without noting that it is positive; this is exactly where $\alpha < \frac{1}{4}\pi$ is used.

Beyond STEP

A cleaner way to see this is to work in the prism's frame, which accelerates at a: there the particle feels an extra "fictitious" force ma backwards, and the problem becomes ordinary statics on a rough slope with an effective gravity tilted by the angle $\arctan(a/g)$. The condition then says that this tilt must lie within λ of α, the angle of friction, which is exactly the interval found. Non-inertial frames and fictitious forces are developed carefully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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